



CSE 641 Computer Vision

Weekly Report 6

Automated Segmentation of Post-Treatment Glioma Sub-regions and Resection Cavities from Multi-Parametric MRI.

Submitted to faculty: Prof. Mehul Raval

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Aim:

To develop and assess deep learning models that can achieve precise and automated 3D brain tumor segmentation on the BraTS dataset from raw MR images.

Introduction:

It is crucial to accurately segment gliomas for surgery planning. Although we have developed a strong baseline model, the SegResNet using a 3D CNN, the localized nature of CNNs is the intrinsic bottleneck. In the past week, we switched gears and explored state-of-the-art Vision Transformers. We used the Swin UNETR model that makes use of self-attention to capture global contextual information from the 3D MRI image. Our objective was to train the Transformer model and compare its performance against the CNN baseline in terms of achieving smooth and contiguous tumor segmentation.

Work Completed:

Swin UNETR Model Integration: Succeeded in incorporating MONAI's Swin UNETR architecture into our current setup. Fixed any environment dependencies (particularly einops needed for complicated tensor manipulations in self-attentive processes).

Vision Transformer Optimization: Considering the weight of vision transformers compared to CNNs, applied gradient checkpointing (use_checkpoint=True) and managed to successfully train the model on an 8 GB GPU without running into OOM error, despite retaining our previous 96x96x96 patch dimensions.

Vision Transformer Training: From scratch, trained Swin UNETR model with PyTorch in Mixed Precision mode (using autocast) to reduce training epoch time, again with the same 80:20 data split to allow for a fair comparison with SegResNet from last week.

Evaluation & Visualization: Executed the enhanced evaluation code using the freshly trained weights to compute the Dice Similarity Coefficient (DSC) and the 95% Hausdorff Distance (HD95). Additionally, we have produced the comparison images for MRI, ground truth, and prediction by the Swin UNETR network.

Results:

The Swin UNETR model successfully learned to segment the tumor sub-regions, demonstrating the viability of self-attention mechanisms for 3D medical imaging. The internal validation results are as follows:

Mean Lesion-wise DSC: 0.7745 (A solid improvement over the CNN baseline of 0.7714).

Mean Lesion-wise HD95: 11.5082 mm (A massive reduction in boundary error compared to the CNN's 16.01 mm, proving the Transformer hallucinates far fewer distant background pixels).

```
warnings.warn(  
✓ Swin UNETR Evaluation Complete!  
Mean Lesion-wise Dice Score (DSC): 0.7745  
Mean Lesion-wise 95% Hausdorff Distance (HD95): 11.5082 mm
```

Clinical Volume Example: Extracted real-world metrics for our sample patient using the Transformer predictions (Tumor Core: 60.22 cm³, Surrounding Edema: 167.97 cm³).

```
Generating Clinical Volume Report using Swin UNETR...  
=====   
● SWIN UNETR CLINICAL VOLUME REPORT ●   
=====   
Tumor Core (ET + NETC) Volume : 60.22 cm3  
Surrounding Edema (SNFH) Vol : 167.97 cm3  
=====
```

Qualitative Observation: Visual outputs indicate that the Swin UNETR produces cohesive segmentations, effectively capturing global tumor context compared to the highly localized predictions of the previous CNN.



Next steps and goals:

Comparison of Models and Model Selection: Perform a thorough comparison of these two baseline models (CNN vs Transformer) quantitatively and qualitatively in terms of their Dice Scores (DSC), Hausdorff Distances (HD95), and boundary smoothness. The final comparison will be documented, and the winner in this comparison will be selected on the basis of the aforementioned metrics.

Hyperparameter Optimization of Selected Model: Once one of the models is selected based on the comparative analysis (the expected winning model is Swin UNETR), fine-tuning of its hyperparameters will commence. It will consist of experimentation with various learning rate schedulers, such as cosine annealing, weight decay, and data augmentation techniques in order to improve the Dice Score.

Integration of Post-Processing Technique: Apply post-processing techniques like "Keep Largest Connected Component" on the output produced by the fine-tuned model in order to eliminate isolated background noise and reduce the HD95 boundary error penalty.

Conclusion:

This week saw the smooth shift from classic Convolutional Neural Networks to Vision Transformers. With the successful implementation and training of the Swin UNETR model, our computational limits have been surpassed, and we successfully tested the segmenting abilities of this new technology. With two baselines trained in both CNN and Transformer models, the latter exhibiting superior qualities in terms of boundary distance measures (HD95), we are ready for some final analytics and post-processing steps next week.

References:

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3. A. Parida et al., "Improving Pre-trained Adult Glioma Segmentation Models using only Post-processing Techniques," arXiv preprint arXiv:2512.14937, 2025.
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